

PHASE 1 — Project Setup & Basic UI

Objective

Setup frontend environment and create basic application structure.

Tasks

Install Frontend Stack

Install:

- React JS
 - Tailwind CSS
 - React Router
 - Axios
 - Chart.js
-

Create Folder Structure

```
frontend/  
|  
├─ src/  
|   ├─ pages/  
|   ├─ components/  
|   ├─ services/  
|   ├─ assets/  
|   └─ layouts/  
|  
└─ public/
```

Setup Routing

Pages:

- Home
- Register
- Attendance

- Dashboard
 - Records
-

Create Global UI Theme

Things to setup:

- colors
 - typography
 - button styles
 - cards
 - spacing
-

Deliverables

- React app setup
 - Tailwind configured
 - Routing working
 - Folder structure ready
-

PHASE 2 — Home Page Development

Objective

Create professional landing page.

Tasks

University Header

Example:

XYZ UNIVERSITY

AI Attendance Management System

Main Buttons

Buttons:

- Register
 - Attendance
-

Responsive Layout

Should work properly on:

- laptop
 - tablet
 - mobile
-

UI Enhancements

Add:

- hover effects
 - transitions
 - icons
-

Deliverables

- Fully working homepage
 - Responsive design
-

PHASE 3 — Registration Page UI

Objective

Create registration workflow UI.

Tasks

Webcam Component

Features:

- webcam preview

- start/stop camera
 - capture button
-

Registration Form

Fields:

- name
 - roll number
 - department
 - semester
 - email
-

Submit Workflow

After capture:

- form opens
 - submit button enabled
-

Success Message UI

Example:

-  Student Registered Successfully
-

Deliverables

- Registration UI completed
 - Webcam integrated
 - Form validation working
-

PHASE 4 — Attendance Page UI

Objective

Create live attendance recognition interface.

Tasks

✓ Attendance Webcam Screen

Features:

- live camera
 - face detection box
 - loading animation
-

✓ Attendance Status Cards

Possible states:

Success

✓ Attendance Marked

Already Marked

⚠ Already Marked

Unknown Face

✗ Unknown Person

✓ Live Attendance Counter

Example:

Present Today: 42

📦 Deliverables

- Attendance UI completed
 - Recognition states displayed properly
-

📌 PHASE 5 — Dashboard Development

Objective

Create admin analytics dashboard.

Tasks

Dashboard Cards

Show:

- total students
 - present students
 - absent students
-

Graphs & Analytics

Charts:

- attendance percentage
 - weekly analytics
 - monthly analytics
-

Student Table

Features:

- search
 - filters
 - pagination
-

Deliverables

- Full admin dashboard
 - Graphs working
-

PHASE 6 — Attendance Records Page

Objective

Create attendance history management screen.

Tasks

Records Table

Columns:

- name
 - roll no
 - date
 - time
 - status
-

Filters

Filter by:

- student
 - date
 - department
-

Export Button UI

Buttons:

- Download Excel
 - Download PDF
-

Deliverables

- Records page completed
-

PHASE 7 — Final UI Polish

Objective

Improve professionalism.

Tasks

 **Dark/Light Mode**

 **Loading Animations**

 **Toast Notifications**

Example:

Attendance Saved

 **Error Screens**

Handle:

- camera permission denied
 - server error
 - no face detected
-

 **Deliverables**

- Final polished frontend